Macroeconomic Drivers, Market Dynamics, and the Predictability of Gold and Silver Prices Under Economic Uncertainty

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Abstract

Gold and silver have long been considered safe-haven assets, particularly during periods of economic instability, inflation, and financial market volatility. Understanding the factors that drive their price movements is important for investors, policymakers, and financial institutions seeking to manage risk and make informed decisions. However, despite their significance, there is limited understanding of how macroeconomic variables and economic policy uncertainty jointly influence the predictability of gold and silver price movements in a rapidly changing global financial environment.
The main objective of this study is to examine how macroeconomic indicators, market variables, and economic policy uncertainty affect gold and silver price dynamics, and to evaluate whether these factors can be used to predict short-term price movements. The study also investigates whether prediction accuracy varies across different levels of economic policy uncertainty.

Data Retrieval

Three data sources were combined into a unified daily panel spanning January 2006 to March 2026.

1. Yahoo Finance (yfinance API) : Daily closing prices for 15+ instruments pulled via yf.download() from Jan 2006 to March 2026. Forward/backward filled for missing trading days. Included Gold, Silver, Crude Oil, S&P 500, Nasdaq, Dollar Index, VIX, Copper, Platinum
2. FRED- Inflation & Interest rate : 5 year breakeven inflation and federal funds rate (DFF). Merged on business day date key with forward fill to align monthly/daily cadences
3. EPU Index : The Economic Policy Uncertainty was captured using the Baker, Bloom & Davis Daily Policy Uncertainty Index.

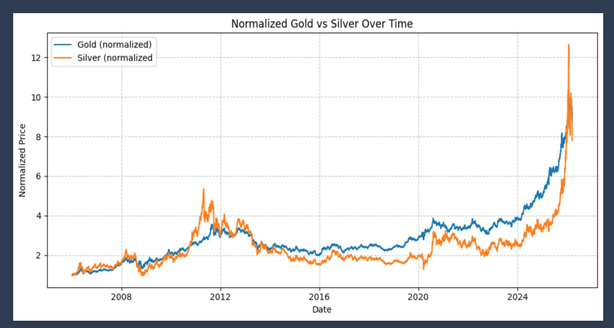
All three datasets were merged on a shared business day date key, with forward filling applied to align monthly or sparse series to the daily frequency

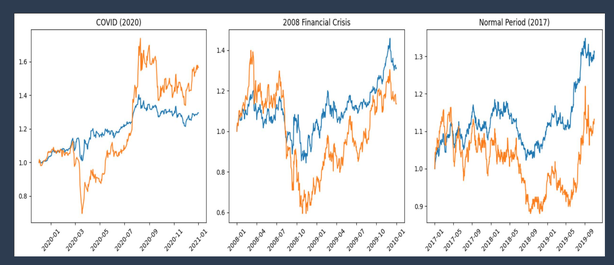
Date	Gold	Silver	Crude Oil	S&P500	Russel 2000 Index	10 Yr US T-Note Futures	2 Yr US T-Note Futures	Platinum	Copper	Dollar Index	...	Euro USD	Euronext100	Nasdaq	inflation_rate	daily_policy_inde
2006-01-02	530.7	9.1	63.1	1268.8	684.0	109.7	102.7	975.8	2.2	89.8	...	1.2	816.6	2243.7	2.30	164.6
2006-01-03	530.7	9.1	63.1	1268.8	684.0	109.7	102.7	975.8	2.2	89.8	...	1.2	820.7	2243.7	2.30	88.6
2006-01-04	533.9	9.1	63.4	1273.5	689.2	109.8	102.8	995.6	2.2	89.1	...	1.2	829.8	2263.5	2.31	61.2
2006-01-05	526.3	8.8	62.8	1273.5	691.8	109.8	102.8	984.5	2.1	89.3	...	1.2	828.6	2276.9	2.28	112.2
2006-01-06	539.7	9.1	64.2	1285.4	699.4	109.6	102.7	996.8	2.2	88.8	...	1.2	833.3	2305.6	2.33	36.3

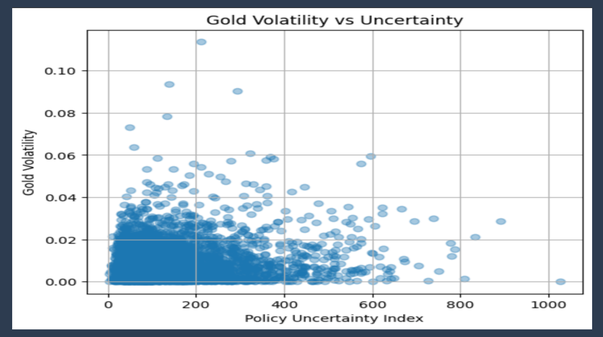
5 rows x 23 columns

Methodology

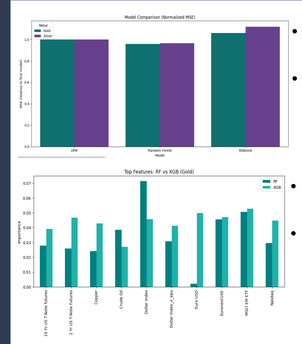
1. Feature Engineering: Daily percentage returns were computed for gold and silver using pct_change() and shifted by one day to create next-day return targets. Three interaction terms were constructed by multiplying EPU with inflation rate, interest rate, and the Dollar Index to capture regime conditional macro effects. A binary high-uncertainty flag (epu_high) was created by thresholding EPU at its 75th percentile, classifying the top 25% of observations as high-uncertainty periods.
2. Models: Three models were trained and compared on a chronological 80/20 train-test split. Linear Regression was trained on standardized features as a baseline. Random Forest used 300 trees with a maximum depth of 8 and minimum 5 samples per leaf. XGBoost used 300 boosting rounds with a learning rate of 0.05 subsampling of 0.8, and column subsampling of 0.8. All models predicted next-day returns for gold and silver separately. Performance was evaluated using MSE, RMSE, MAE, and R2
3. Crisis Regime Tests: Two out-of-sample regime experiments were conducted. For the 2008 Financial crisis, models were trained on data through December 2007 and tested on January 2008 to January 2010. For COVID-!9, models were trained on January 2017 to December 2019 and tested on December 2019 to January 2022. Linear regression coefficients were compared across the full dataset and each crisis period to identify structural shifts in macro-variable relationship







Results -Part 1



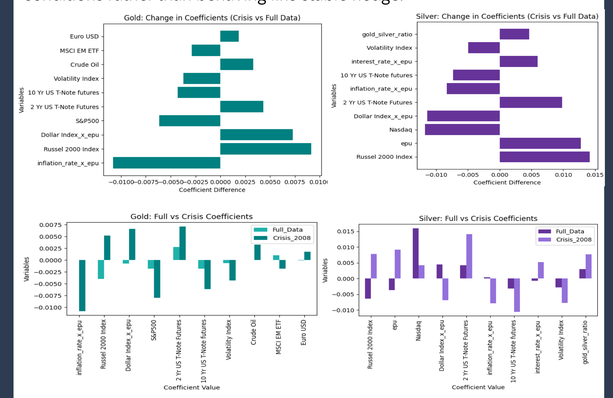
- Each model is normalized allowing comparison across gold and silver.
- Random Forest performs the best across both gold and silver. XGBoost shows highest error across both gold and silver.
- Random forest places the highest importance on 'Dollar Index'
- XGBoost distributes importance more evenly across multiple features like MSCI EM ETF, Nasdaq and Euro USD.

Results - Part 2

Impact of Financial Recession

Gold: Relationship with macro variables shift significantly, especially stronger negative links with equities under uncertainty. Shows that gold acts as a clearer safe-haven during financial stress.

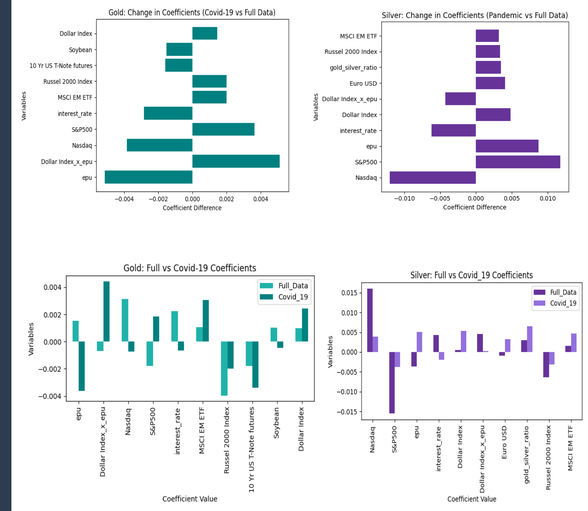
Silver: Larger and more unstable coefficient swings with stronger reversals, indicates that silver becomes highly sensitive to market conditions rather than behaving like stable hedge.



Impact of Covid-19

Gold: Coefficient changes are present but less extreme than 2008, suggesting gold hedges uncertainty but with more mixed and less consistent macro relationships

Silver: Continues to show volatile and inconsistent shifts, reinforcing that it reacts more to short-term market movements and uncertainty.



Conclusion

1. Daily gold and Silver returns proved unpredictable across all three models at the daily frequency. Market indicators like the Dollar Index, VIX, and gold-silver ratio were more informative than traditional macro fundamentals like inflation and interest rates whose lagged and nonlinear effects are difficult to capture at daily level.
2. Economic policy uncertainly does not improve prediction but worsens it. High EPU periods amplified model error across all approaches, suggesting than uncertainty disrupts macro-price relationships rather than creating exploitable signals. Crisis periods confirmed this instability: the 2008 crisis and COVID-19 both produced structural breaks in model coefficients, with gold acting as reliable safe haven while silver's industrial exposure caused sharper initial declines and faster recoveries.
3. Future work could explore regime-switching models, LSTMs for temporal dependencies, or rolling-window training to better adapt to shifting market conditions.